

Automated Melanoma Skin Cancer Detection and Interpretability Analysis via Transfer Learning, Clinical Threshold Calibration, and Grad-CAM

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Abstract—Cutaneous melanoma is the most lethal form of skin malignancy, accounting for over 75% of skin cancer mortalities despite constituting under 5% of total skin neoplasm incidence. Early clinical intervention yields five-year survival rates exceeding 98%, emphasizing the necessity of automated, high-sensitivity computer-aided triaging systems. In this work, an end-to-end deep learning framework is developed for the binary classification of dermoscopic lesions into benign nevi and malignant melanoma using a benchmark cohort of 13,879 images (224×224 pixels). An empirical ablation study investigates a custom 3-block convolutional neural network (CNN) trained from scratch ($\sim 94K$ parameters) against transfer learning adaptations of ResNet-18 (linear probing and deep residual fine-tuning) and EfficientNet-B0. To prioritize clinical safety and alleviate dangerous false negatives, an empirical decision threshold calibration sweep ($\tau \in [0.20, 0.50]$) is coupled with a weighted soft-voting ensemble ($0.60R + 0.40E$). Evaluated on a held-out test cohort of 2,000 images, fine-tuned ResNet-18 achieves 94.05% accuracy, 93.50% sensitivity, and 0.9858 ROC-AUC at default threshold $\tau = 0.50$. Calibrating to $\tau = 0.35$ increases clinical sensitivity to 96.00%, reducing missed malignancies from 65 to 40. Furthermore, the soft-voting ensemble demonstrates superior performance, attaining 95.20% accuracy and 0.9903 ROC-AUC at $\tau = 0.50$, while calibration at $\tau = 0.35$ raises sensitivity to 97.20%, achieving an 84.40% reduction in missed melanomas (28 false negatives vs. 180 in the baseline). Qualitative validation via hook-based Gradient-weighted Class Activation Mapping (Grad-CAM) verifies that network inferences are driven by genuine dermatological lesion morphology (pigment network asymmetry, border irregularity) rather than spurious background artifacts.

Index Terms—Computer Vision, Dermoscopy, Melanoma Detection, Transfer Learning, Clinical Threshold Calibration, Explainable AI, Grad-CAM.

I. INTRODUCTION

CUTANEOUS malignant melanoma represents one of the most aggressive human malignancies, characterized by rapid metastatic dissemination via lymphatic and hematogenous routes. Although melanoma constitutes less than 5% of all diagnosed skin cancers, it is responsible for more than 75% of all skin cancer-related mortalities globally [1]. As prognostic outcomes are directly governed by tumor thickness at detection

(Breslow depth), the prompt management of thin, localized Stage I lesions (<0.8 mm) yields 5-year relative survival rates exceeding 98%. In contrast, regional nodal metastasis diminishes survival to 64%, dropping precipitously below 23% once distant visceral metastasis occurs.

Dermoscopy, which is a non-invasive epiluminescence optical microscopy technique that eliminates surface skin reflection via liquid immersion or cross-polarization, substantially enhances visual differentiation of subsurface pigmented structures. Nonetheless, manual diagnostic sensitivity among general clinical practitioners ranges between 60% and 75%, achieving 85% - 90% only under specialized dermatological expertise [2]. Developing automated, reproducible, and objective computer vision triaging algorithms is therefore vital to bridge diagnostic disparity.

A. The Asymmetric Cost of Diagnostic Misclassification

Conventional machine learning classifiers are typically trained and evaluated using symmetric loss objectives (such as standard accuracy or cross-entropy loss), which penalize False Positives (FP) and False Negatives (FN) equally. In an oncological screening context, however, misclassification penalties are fundamentally asymmetric.

- **False Positive (Type I Error):** A benign nevus is predicted as malignant, resulting in non-critical consequences such as secondary specialist referral or a minor diagnostic biopsy.
- **False Negative (Type II Error):** A life-threatening melanoma is misdiagnosed as benign, leading to delayed medical intervention, disease progression, and avoidable patient mortality.

Consequently, computer-aided dermoscopy systems must prioritize **Sensitivity (Recall)** and **Area Under the ROC Curve (ROC-AUC)** over symmetric accuracy, necessitating rigorous empirical calibration of classification decision thresholds.

B. Contributions of this Work

To address these clinical and computational requirements, this paper provides the following contributions:

- 1) **Empirical Baseline vs. Transfer Learning Ablation:** A custom 3-block CNN ($\sim 94K$ parameters) is constructed from first principles and compared against transfer learning adaptations of ResNet-18 (feature extraction vs. deep residual fine-tuning) and EfficientNet-B0 on a

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Course: CO5430 Image Processing Project (Medical Imaging Track), July-September 2026.

Source Code and Reproducibility Repository: <https://github.com/cepdnacl/e22-co5430-melanoma-cancer-detection>

standardized cohort of 13,879 images at native 224×224 resolution.

- 2) **Clinical Threshold Calibration Study:** A systematic sensitivity analysis across decision thresholds $\tau \in [0.20, 0.50]$ demonstrates that shifting the threshold to $\tau = 0.35$ reduces missed cancers by 38.5% with negligible loss in specificity.
- 3) **Weighted Soft-Voting Ensemble (Stretch Goal):** A dual-backbone probability ensemble ($0.60R + 0.40E$) attains 95.20% accuracy and 0.9903 ROC-AUC, reducing false negatives from 180 (baseline) down to 28 at $\tau = 0.35$ (97.20% sensitivity).
- 4) **Explainable AI (Grad-CAM) Validation:** Hook-based Gradient-weighted Class Activation Mapping is implemented to qualitatively audit spatial attention across four clinical prediction categories (Correct Benign, Correct Malignant, False Positive, False Negative) and contrast feature representations between scratch and pretrained models.
- 5) **Reproducible Open-Source Package:** Complete release of code, trained model checkpoints, and structured metrics ("metrics.json") under an open MIT license.

II. RELATED WORK

A. Classical Feature Extraction in Dermoscopy

Prior to deep representation learning, automated skin lesion analysis relied on hand-crafted mathematical descriptors mimicking the clinical **ABCDE criteria** (Asymmetry, Border irregularity, Color variegation, Diameter $> 6\text{mm}$, Evolution). Border irregularity was modeled using compactness indices, fractal dimensions, and Fourier boundary harmonics. Color variegation was quantified via color histogram moments, and texture was parameterized using Gray-Level Co-occurrence Matrices (GLCM) and Local Binary Patterns (LBP). Classification was traditionally conducted using Support Vector Machines (SVM) or Random Forests. While computationally lightweight and interpretable, hand-engineered pipelines suffered from extreme sensitivity to illumination variability, skin tone discrepancies, and non-lesion artifacts (e.g., hair occlusions, gel bubbles).

B. Deep Convolutional Neural Networks and Transfer Learning

The landmark work of Esteva *et al.* [1] proved that deep convolutional networks (Inception-v3) pretrained on generic images and fine-tuned on clinical datasets could match the diagnostic consensus of 21 board-certified dermatologists. Subsequent works established residual networks (ResNet) [3] and compound-scaled architectures (EfficientNet) [4] as dominant backbones. Transfer learning leverages generic spatial primitives (edges, textures, color gradients) learned from ImageNet, mitigating the severe overfitting risks associated with training deep networks on medical cohorts from scratch.

C. Visual Interpretability and Explainable AI

Because deep neural networks operate as non-linear black boxes, clinicians hesitate to deploy systems lacking transparent

decision rationale. Selvaraju *et al.* [5] introduced Gradient-weighted Class Activation Mapping (Grad-CAM), utilizing convolutional gradients to construct coarse 2D saliency maps that highlight discriminative spatial regions. In medical imaging, Grad-CAM serves as an indispensable audit mechanism to detect "shortcut learning", such as models classifying lesions based on surgical skin markings or lens vignetting rather than actual pathological structures.

III. DATASET AND PREPROCESSING

A. Cohort Overview and Partitioning

This project utilizes the Kaggle Melanoma Cancer Dataset, comprising 13,879 dermoscopic images organized into binary classes: "Benign" ($n = 7,289$, 52.5%) and "Malignant" ($n = 6,590$, 47.5%). To prevent data leakage and provide unbiased generalization benchmarks, the cohort is partitioned into three disjoint splits.

- **Training Partition:** 10,691 images (77.03%) used for gradient updates.
- **Validation Partition:** 1,188 images (8.56%) used for early stopping and model selection.
- **Held-Out Test Cohort:** 2,000 images (1,000 Benign, 1,000 Malignant; 14.41%) strictly reserved for final quantitative reporting.

B. Channel Statistics and Normalization

Rather than assuming generic natural distributions, we compute the empirical per-channel RGB mean (μ) and standard deviation (σ) across the training cohort.

$$\begin{aligned}\mu_{\text{dataset}} &= [0.7635, 0.5461, 0.5704] \\ \sigma_{\text{dataset}} &= [0.1409, 0.1519, 0.1695]\end{aligned}\tag{1}$$

The baseline CNN trained from scratch uses these empirical statistics. For transfer learning architectures initialized from ImageNet-1K weights, inputs are normalized using standard ImageNet parameters ($\mu_{\text{ImageNet}} = [0.485, 0.456, 0.406]$, $\sigma_{\text{ImageNet}} = [0.229, 0.224, 0.225]$). All inputs are processed at 224×224 pixels.

C. Data Augmentation Pipeline

To enforce spatial invariance and prevent memorization of lesion orientations, stochastic transformations are applied during training.

- Random horizontal reflections ($p = 0.5$)
- Random vertical reflections ($p = 0.2$)
- Stochastic affine rotations ($\theta \in [-15^\circ, +15^\circ]$)
- Color jitter (Brightness $\pm 10\%$, Contrast $\pm 10\%$, Saturation $\pm 10\%$)

Validation and test samples undergo strictly deterministic tensor conversion and channel normalization.

IV. METHODOLOGY AND ARCHITECTURES

A. Custom Baseline CNN (from Scratch)

To establish an empirical benchmark, a 3-block convolutional neural network was implemented from scratch. Each building block comprises:

$$\text{Conv2D}(3 \times 3) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPool2D}(2 \times 2) \quad (2)$$

Channel dimensions expand as $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$. Spatial feature maps are collapsed using Global Average Pooling (GAP) into a 128-dimensional latent vector, followed by Dropout ($p = 0.4$), a dense linear layer ($128 \rightarrow 64$), ReLU activation, Dropout ($p = 0.2$), and a single linear output logit. Total trainable parameters equal **94,337**.

B. ResNet-18 (Feature Extraction vs. Fine-Tuning)

ResNet-18 incorporates residual shortcut connections to bypass vanishing gradients:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (3)$$

We examine two adaptation strategies:

- 1) **Feature Extraction (Linear Probing):** The entire convolutional backbone is frozen. The 1,000-class output layer is replaced with "Dropout($p=0.3$) $\rightarrow$ Linear(512, 1)". Only 513 parameters are optimized.
- 2) **Fine-Tuning:** Shallow stages ("conv1", "layer1", "layer2") remain frozen to retain generic edge/texture primitives. Deep residual stages ("layer3", "layer4") and the custom head are unfrozen ($\sim 4.7\text{M}$ parameters) and optimized at a reduced learning rate ($\eta = 3 \times 10^{-4}$).

C. EfficientNet-B0 (Compound Scaling)

EfficientNet-B0 optimizes network width (w), depth (d), and input resolution (r) concurrently via a compound scaling coefficient ϕ :

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi \quad \text{s. t.} \quad \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \quad (4)$$

Utilizing Mobile Inverted Bottleneck Convolutions (MBConv) with squeeze-and-excitation attention, the final feature blocks ("features[7]", "features[8]") and classifier ("Dropout(0.3) $\rightarrow$ Linear(1280, 1)") are fine-tuned ($\sim 4.0\text{M}$ parameters).

D. Weighted Soft-Voting Ensemble

To exploit complementary feature representations and minimize single-model variance, posterior class probabilities are combined via weighted soft voting.

$$P_{\text{ens}}(\text{Malignant}) = 0.60 \cdot P_{\text{ResNet18}} + 0.40 \cdot P_{\text{EfficientNetB0}} \quad (5)$$

The 0.60/0.40 weighting reflects ResNet-18's empirical superiority in baseline sensitivity.

E. Optimization Protocol

All architectures are trained using numerically stable Binary Cross-Entropy with Logits.

$$\mathcal{L}_{\text{BCE}}(z, y) = \max(z, 0) - z \cdot y + \log(1 + e^{-|z|}) \quad (6)$$

where z is the unscaled logit and $y \in \{0, 1\}$ is the ground-truth label.

- **Optimizer:** AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$) [8].

- **Learning Rate Schedule:** Cosine Annealing:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{t}{T_{\max}} \pi \right) \right) \quad (7)$$

with $\eta_{\min} = 10^{-6}$.

- **Batch Size:** 32.
- **Epoch Allocation:** 20 epochs for Baseline CNN (to guarantee full loss stabilization from random initialization) and 15 epochs for transfer learning backbones.
- **Checkpointing:** Model weights are saved at the epoch of minimal validation loss.

V. EXPERIMENTAL RESULTS AND ABLATION

A. Evaluation Metrics

Models are evaluated across seven statistical metrics on the 2,000 held-out test images:

- **Accuracy:** $(\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$
- **Sensitivity (Recall):** $\text{TP} / (\text{TP} + \text{FN})$ (Clinical Safety)
- **Specificity:** $\text{TN} / (\text{TN} + \text{FP})$
- **Precision:** $\text{TP} / (\text{TP} + \text{FP})$
- **F1-Score:** $2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$
- **ROC-AUC:** Area under the ROC curve across continuous thresholds.
- **PR-AUC:** Area under the Precision-Recall curve.

B. Quantitative Analysis

As documented in Table I:

- 1) **Impact of Transfer Learning:** Fine-tuning ResNet-18 increases accuracy from 86.50% to 94.05% and sensitivity from 82.00% to 93.50% compared to the scratch baseline, reducing missed cancers from 180 to 65.
- 2) **Feature Extraction vs. Fine-Tuning:** Frozen feature extraction yields 88.05% accuracy, whereas fine-tuning deep layers unlocks a 6.00% accuracy boost, proving that high-level dermoscopic texture requires domain-specific weight updates.
- 3) **Ensemble Superiority:** The soft-voting ensemble achieves peak discriminative performance across all metrics (95.20% accuracy, 0.9903 ROC-AUC), successfully filtering individual backbone prediction errors.

VI. CLINICAL DECISION THRESHOLD CALIBRATION

Operating diagnostic classifiers at default symmetric thresholds ($\tau = 0.50$) is suboptimal for life-critical conditions. We conduct a systematic empirical threshold calibration sweep on ResNet-18 Fine-Tuned across $\tau \in [0.20, 0.50]$, as presented in Table II.

TABLE I
COMPREHENSIVE EXPERIMENTAL ABLATION STUDY ON 2,000 HELD-OUT TEST IMAGES

Model Architecture	Strategy	Threshold (τ)	Accuracy	Sensitivity	Specificity	Precision	F1-Score	ROC-AUC	Missed (FN)
Baseline CNN	Trained from Scratch	0.50	86.50%	82.00%	91.00%	0.9011	0.8586	0.9461	180
ResNet-18	Feature Extraction	0.50	88.05%	85.80%	90.30%	0.8984	0.8777	0.9478	142
EfficientNet-B0	Fine-Tuned Backbone	0.50	93.85%	91.70%	96.00%	0.9582	0.9371	0.9846	83
ResNet-18	Fine-Tuned (Layers 3+4)	0.50	94.05%	93.50%	94.60%	0.9454	0.9402	0.9858	65
ResNet-18 (Calibrated)	Fine-Tuned (Layers 3+4)	0.35	94.55%	96.00%	93.10%	0.9329	0.9463	0.9858	40
Ensemble Model	0.60R + 0.40E	0.50	95.20%	94.30%	96.10%	0.9603	0.9516	0.9903	57
Ensemble (Best Safety)	0.60R + 0.40E	0.35	95.05%	97.20%	92.90%	0.9319	0.9515	0.9903	28

TABLE II
DECISION THRESHOLD CALIBRATION SWEEP (RESNET-18 FINE-TUNED)

Threshold (τ)	Accuracy	Sensitivity	Specificity	F1	FN	FP
0.20	93.80%	97.70%	89.90%	0.9403	23	101
0.25	94.10%	97.00%	91.20%	0.9427	30	88
0.30	94.30%	96.40%	92.20%	0.9442	36	78
0.35 (Optimal)	94.55%	96.00%	93.10%	0.9463	40	69
0.40	94.50%	95.50%	93.50%	0.9455	45	65
0.45	94.35%	94.60%	94.10%	0.9436	54	59
0.50 (Default)	94.05%	93.50%	94.60%	0.9402	65	54

A. Clinical Implications of Threshold Calibration

Calibrating τ from 0.50 $\rightarrow$ 0.35:

- Increases Sensitivity from 93.50% to **96.00%**.
- Reduces False Negatives from 65 to **40** (a 38.5% reduction in missed cancers).
- Concurrently improves overall accuracy (94.05% $\rightarrow$ 94.55%) and F1-score (0.9402 $\rightarrow$ 0.9463).
- Applied to the **Ensemble Model**, missed melanomas drop to **only 28** out of 1,000 (Sensitivity = **97.20%**), achieving an overall **84.4% reduction in missed cancers** compared to the baseline CNN (180 FN $\rightarrow$ 28 FN).

VII. EXPLAINABLE AI VIA GRAD-CAM

A. Mathematical Formulation

To verify visual reasoning, hook-based Grad-CAM [5] is integrated into the final convolutional layer ("layer4[-1].conv2" of ResNet-18). Neuron importance weights α_k^c for class c (prior to sigmoid) are derived via spatial global average pooling:

$$\alpha_k^c = \frac{1}{Z} \sum_{i=1}^U \sum_{j=1}^V \frac{\partial y^c}{\partial A_{i,j}^k} \quad (8)$$

where A^k denotes the k -th feature map of dimensions $U \times V$, and $Z = U \times V$. The class-discriminative localization map $L_{\text{Grad-CAM}}^c$ is computed by taking a rectified linear combination:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (9)$$

The coarse map is bilinearly upsampled to 224×224 and superimposed onto the original dermoscopic image using a JET colormap overlay ($\alpha = 0.45$).

B. Qualitative Analysis Across Clinical Categories

Grad-CAM heatmaps generated across the four clinical prediction categories demonstrate distinct spatial characteristics.

- **Correct Malignant (True Positives):** Saliency maps exhibit dense focal activations concentrated strictly over asymmetrical pigment networks and irregular notched margins.
- **Correct Benign (True Negatives):** Heatmaps display uniform, low-magnitude activations distributed evenly across symmetric nevi.
- **False Positives:** Activations are triggered by atypical dysplastic nevi with dark central dots, hair occlusions, or peripheral lighting vignetting.
- **False Negatives:** Faint activations occur in amelanotic (hypopigmented) lesions that lack dark melanin pigmentation against fair background skin.

C. Scratch Baseline vs. ResNet-18 Localization

On disagreement cases where models predict opposing labels, the baseline CNN produces diffuse, noisy saliency maps that frequently bleed into healthy surrounding tissue. In contrast, fine-tuned ResNet-18 demonstrates sharp spatial containment, confirming that transfer learning fosters biologically relevant feature grounding.

VIII. FAILURE CASES, LIMITATIONS, AND ETHICS

A. Failure Mode Diagnostics

- **Amelanotic Melanoma:** Lesions deficient in melanin represent the primary source of false negatives, as convolutional kernels rely on pigmentation contrast gradients.
- **Hair Artifacts:** Dense dark hair traversing pale lesions induces high-frequency edge responses, occasionally inducing false positives.

B. Limitations

- **Binary Constraint:** Clinical dermoscopy involves complex multi-class differential diagnosis (e.g.: Basal Cell Carcinoma, Seborrheic Keratosis).
- **Absence of Clinical Metadata:** Patient age, anatomical location, family history, and diameter were omitted due to dataset constraints.

C. Ethics, Safety, and AI Tool Declaration

- **Medical Safety Disclaimer:** This software is designed strictly for academic research under the University of Peradeniya curriculum. It is not an approved medical device and must never substitute for biopsy or clinical judgment.
- **Ethics and De-identification:** The cohort utilizes de-identified public dermoscopy images; no protected personal health information was accessed.
- **AI Tool Usage:** Generative AI tools were used for code syntax debugging and LaTeX formatting. Additionally, similar AI tools were utilized to understand the Grad-CAM concept and guide its implementation. All architectural implementations, training, and analysis were conducted by the student authors.

IX. CONCLUSION AND FUTURE WORK

This work developed an end-to-end medical computer vision framework for automated melanoma detection. Progressing from an empirical scratch baseline (86.50% accuracy) through transfer learning (ResNet-18 and EfficientNet-B0) to a dual-backbone ensemble, the system achieved 95.20% accuracy and 0.9903 ROC-AUC on 2,000 unseen test samples. By calibrating the decision threshold to $\tau = 0.35$, missed melanomas dropped from 180 to 28 (97.20% sensitivity). Grad-CAM confirmed that model predictions align with clinical dermatological features rather than imaging artifacts. Future extensions will incorporate automated hair removal (DullRazor), multimodal patient metadata integration, and multi-class classification.

ACKNOWLEDGMENT

The authors express sincere gratitude to Dr. Upul Jayasinghe, Department of Computer Engineering, University of Peradeniya, for guidance, technical insight, and supervision throughout this project.

INDIVIDUAL CONTRIBUTION STATEMENT

In accordance with Section 8 of the CO5430 Project Specification, the individual contributions of all group members are summarized in Table III.

TABLE III
INDIVIDUAL MEMBER CONTRIBUTION BREAKDOWN

Reg. No.	Member Name	Primary Technical Responsibilities	Share
E/22/052	K. H. D. M. Bimsara	Dataset acquisition, EDA, RGB normalization statistics, baseline CNN design & training.	25%
E/22/058	M. M. T. Cooray	ResNet-18 (FE & FT), cosine annealing scheduler, training pipeline optimization.	25%
E/22/353	G. K. G. Sandeepa	EfficientNet-B0 fine-tuning, ensemble blending, threshold calibration study.	25%
E/22/419	R. G. S. T. Weerasekara	Hook-based Grad-CAM engine, saliency grids, error analysis, repo maintenance, LaTeX draft.	25%

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